Module 7

Capital Budgeting

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

Goals

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- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept

Roadmap

Should we invest in a project (e.g., a data center)?

NPV > 0?

Recall from **Module 6**

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How to forecast cash flows?

- Similar to **Module 2**, indirect method for cash flow
- But different, cash flow from a project, not a whole firm

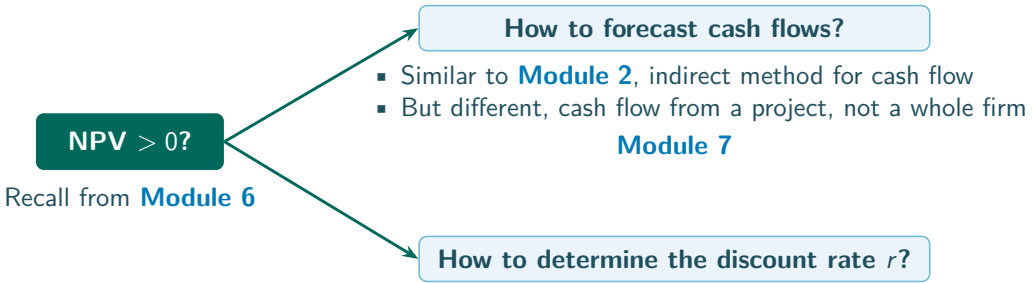
Module 7

NPV > 0?

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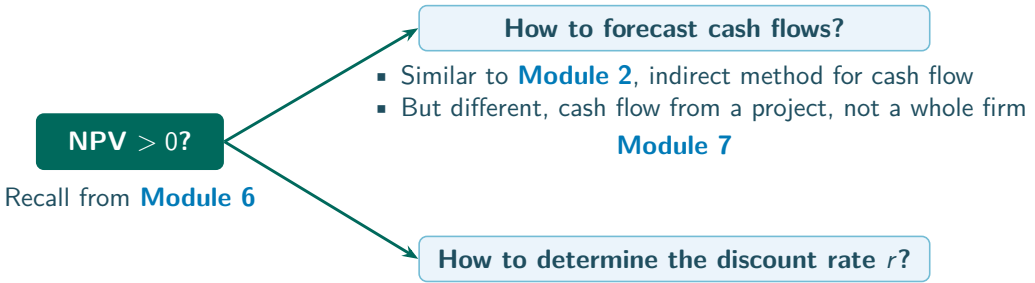
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Motivating Example, Is AI Spending a Bubble?



CNBC, August 12, 2026 (7:21), "We're 'absolutely' in an AI bubble right now, says Lead Edge Capital's Mitchell Green"

Module 7, Part 0

Review of the Statement of Cash Flows

Statement of Cash Flows

Recall from **Module 2**

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made

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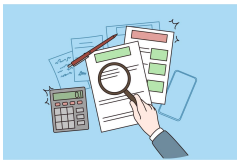
Recall from **Module 2**

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	**The common choice**

Direct →



Indirect →



Statement of Cash Flows

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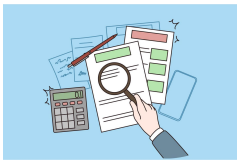
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Statement of Cash Flows

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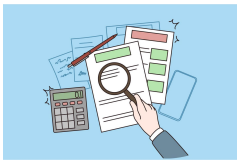
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



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Cash flow (\$ billions)	**−2.0**	0	0	0	0	0	
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Relationship of cash flow (CF) and net income (NI)?
When spending on fixed assets, CF = NI − spending on fixed assets
When recording depreciation, CF = NI + depreciation

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Recall from **Module 2**

- The firm spends $60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

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Recall from **Module 2**

- The firm owed suppliers $30 million for electronic parts
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When firm pays off suppliers, CF = NI + change in accounts payable

Operating Activities

Recall from **Module 2**

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

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$$CF = NI - \text{change in net working capital (NWC)}$$

Investing Activities

Recall from **Module 2**

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

Financing Activities

Recall from **Module 2**

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Recap of Part 0

Recall from **Module 2**

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow = Net income + Depreciation − Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Module 7, Part 1

Project Cash Flow

Firm Cash Flow vs. Project Cash Flow

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Firm Cash Flow vs. Project Cash Flow

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- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC
- **Investing activities**, the project's capital investments, buying its assets at the start and selling them at the end
- **Financing activities**, *none*, we do not consider financing activities when evaluating a project on its own

Identify Project Cash Flow

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$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

Component 1, Project Net Income

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Revenues from the project
– COGS and SG&A expenses for the project
– Depreciation for the project's capital investment
Project EBIT
– Taxes (tax rate $\times$ project EBIT)
Project net income

Component 2, Depreciation

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- To get cash flow from the project, add back depreciation, as in **Module 2**

Example, Project Net Income and Depreciation

- A project requires an investment in fixed assets of $2,000. The project's revenues minus its COGS and SG&A expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%

Example, Project Net Income and Depreciation

- A project requires an investment in fixed assets of $2,000. The project's revenues minus its COGS and SG&A expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%
- Calculate the project's net income plus depreciation for year 1

Example, Project Net Income and Depreciation

- Solution,

Item	$
Revenues − COGS − SG&A expenses	1,100
− Depreciation	1,000
Project EBIT	**100**
− Taxes	21
Project net income	**79**
+ Depreciation	1,000
Project net income + Depreciation	**1,079**

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- Solution,

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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$

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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$
- Project net income + Depreciation $= 79 + 1{,}000 = \$1{,}079$

Component 3, Cash Flow from Changes in NWC

- Recall from **Module 2**,
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- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow,

Cash flow from changes in NWC = −Change in the project's NWC

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Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	$+100$	0	0	-100
Cash flow from changes in NWC	-100	0	0	$+100$

Component 3, Cash Flow from Changes in NWC

- E.g., a 3-year project needs \$100 of NWC while it operates, recovered when the project ends

Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	$+100$	0	0	-100
Cash flow from changes in NWC	-100	0	0	$+100$

- The \$100 goes out at $t = 0$ and returns only at $t = 3$, so tying up the cash still costs the time value of money

Component 4, Cash Flow from Capital Investments

- If the project requires its own investment in fixed assets to get started (e.g., building a data center), that is a negative cash flow

Component 4, Cash Flow from Capital Investments

- If the project requires its own investment in fixed assets to get started (e.g., building a data center), that is a negative cash flow
- Fixed assets are usually depreciated "straight-line" over time,
  - **Expected life**, the number of years the project uses the fixed assets

$$\text{Annual depreciation} = \frac{\text{Capital investment}}{\text{Expected life}}$$

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Recap of Part 1

- Project net income comes from the project's own income statement, revenues less COGS and SG&A expenses, depreciation, and taxes
- Depreciation is not a cash flow, add it back to the project's net income
- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow
- The investment in fixed assets is a negative cash flow, and the fixed assets are depreciated straight-line over the expected life

Module 7, Part 2

Putting It All Together

Example, Putting It All Together

- A company is considering building a new data center, an investment in fixed assets of $1 billion ($1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years

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- A company is considering building a new data center, an investment in fixed assets of \$1 billion (\$1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years
- The data center needs an increase in NWC of \$100 million at $t = 0$ (spare servers and parts), recovered fully at $t = 5$

Example, Putting It All Together

- A company is considering building a new data center, an investment in fixed assets of \$1 billion (\$1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years
- The data center needs an increase in NWC of \$100 million at $t = 0$ (spare servers and parts), recovered fully at $t = 5$
- Renting out its computing capacity brings in \$240 million a year after COGS and SG&A expenses (electricity, staff, and so on)

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- Renting out its computing capacity brings in \$240 million a year after COGS and SG&A expenses (electricity, staff, and so on)
- The corporate tax rate is 21%, and the discount rate is 8%
- Should the company build the data center or not?

Identify Project Cash Flow

- Cash flow from a project has four components,

$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5

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Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240

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- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200

- Depreciation $= 1{,}000/5 = 200$ a year

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**

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− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4

- Depreciation $= 1{,}000/5 = 200$ a year
- Taxes $= 21\% \times 40 = 8.4$

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Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**

- Depreciation $= 1{,}000/5 = 200$ a year
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Project net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		200	200	200	200	200

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5
Project net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**

- The cash for the data center went out at $t = 0$, it is counted in Component 4

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5

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The project's NWC	100	100	100	100	100	0

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- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

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The project's NWC	100	100	100	100	100	0
Change in NWC	+100	0	0	0	0	−100

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5
The project's NWC	100	100	100	100	100	0
Change in NWC	+100	0	0	0	0	−100
Cash flow from changes in NWC	**−100**	**0**	**0**	**0**	**0**	**+100**

- An increase in NWC is a negative cash flow, and a decrease is a positive cash flow

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5

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- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5
Investment in fixed assets	−1,000	0	0	0	0	0

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5
Investment in fixed assets	−1,000	0	0	0	0	0
Cash flow from capital investments	**−1,000**	**0**	**0**	**0**	**0**	**0**

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

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Cash flow from changes in NWC	−100	0	0	0	0	+100

Example, Total Cash Flows

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Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0
Total cash flows	**−1,100**	**231.6**	**231.6**	**231.6**	**231.6**	**331.6**

- At $t = 0$, $-1{,}000 - 100 = -1{,}100$. At $t = 5$, $231.6 + 100 = 331.6$

Example, NPV and the Decision

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$$\text{NPV} = -1{,}100 + \frac{231.6}{1.08} + \frac{231.6}{1.08^2} + \frac{231.6}{1.08^3} + \frac{231.6}{1.08^4} + \frac{331.6}{1.08^5} = -107.23$$

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- NPV < 0, so the company should not build the data center

Example, NPV with the Cash Flow Worksheet

One cash flow at a time

Enter	Keys
	CF, **2nd**, **CLR Work**
−1100	**ENTER**, ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
331.6	**ENTER**
	NPV (the calculator asks for I)
8	**ENTER**, ↓
	CPT
−107.23	

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231.6	ENTER, ↓ ↓
331.6	ENTER
	NPV (the calculator asks for I)
8	ENTER, ↓
	CPT
−107.23	

Repeated cash flow with frequency

Enter	Keys
	CF, 2nd, CLR Work
−1100	ENTER, ↓
231.6	ENTER, ↓
4	ENTER (the frequency F01), ↓
331.6	ENTER
	NPV (the calculator asks for I)
8	ENTER, ↓
	CPT
−107.23	

Module 7, Part 3

The Principles behind Project Cash Flows

General Principles for Identifying Project Cash Flow

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1. Discount cash flows, not net income
2. Ignore cash flows from financing activities
3. Don't forget about depreciation and taxes
4. Only consider incremental cash flows

Principle 1, Discount Cash Flows, Not Net Income

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- Differences in how capital expenditures, depreciation, and NWC are handled
- We always discount the actual cash flows to their present value
- We do not discount net income

Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year

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- The discount rate is 10%

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- Compare the NPV using net income to the NPV using cash flows

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Clearly a money-loser!

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- Solution,

Year	0	1	2

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Revenues − COGS − SG&A expenses		1,500	500
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- Solution,

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− Depreciation		1,000	1,000
Net income		**500**	**−500**

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
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Net income		**500**	**−500**
+ Depreciation		1,000	1,000
Net income + Depreciation	**0**	**1,500**	**500**

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Year	0	1	2
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Net income		**500**	**−500**
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Net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
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Net income + Depreciation	**0**	**1,500**	**500**
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Net income		**500**	**−500**
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Example, Net Income vs. Cash Flows

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Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Net income		**500**	**−500**
+ Depreciation		1,000	1,000
Net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project
- NPV using cash flows $= -2{,}000 + \frac{1{,}500}{1.10} + \frac{500}{1.10^2} = -\$223.14$
- NPV < 0, so the project should be rejected, net income gets it wrong

Principle 2, Ignore Cash Flows from Financing Activities

- Capital budgeting focuses on the investment decision
  - Do not conflate it with the financing decision of how to pay for the project (cash on hand, new debt, or new equity)

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  - In the examples, no interest is subtracted when calculating the project's cash flows, and no interest tax shield is counted
  - The effects of interest (and the interest tax shield) enter when we *separately* consider financing decisions
- The data center table has no interest line, even if the company borrows to build it

Principle 3, Don't Forget about Depreciation and Taxes

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- Depreciation is not a cash flow, but it lowers taxable income and so the taxes the project pays, the depreciation tax shield
- Different depreciation schedules, in this course we stick to "straight-line" depreciation, as in the data center, $1{,}000/5 = \$200$ million a year

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$$\text{Incremental cash flow} = \text{Cash flow with the project} - \text{Cash flow without it}$$

- In the data center table, the rental income and the spare parts exist only because of the data center, so they count

Incremental Cash Flows, a Few Rules That Are Easy to Forget

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f. Remember terminal cash flows

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- Nintendo plans a new-generation gaming console. It requires an investment of $200m and produces cash flows of $400m and $800m in years 1 and 2. Looks like a great investment!





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- But the launch will hurt sales of the older console, forecast to create $800m and $1,000m in years 1 and 2, only $500m and $400m with the new one on the market





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- But the launch will hurt sales of the older console, forecast to create \$800m and \$1,000m in years 1 and 2, only \$500m and \$400m with the new one on the market
- What is the **incremental** cash flow of the new console?





Rule a, Indirect Effects

- Solution ($ millions),

	$t = 0$	$t = 1$	$t = 2$
Cash flow from the old console without the new one		800	1,000
Cash flow from the old console with the new one		500	400
Cash flow from investment in and sales of the new one	-200	400	800
Incremental cash flow of the new console	-200	$400 - 300 = 100$	$800 - 600 = 200$

Rule b, Sunk Costs

- **Sunk costs**, expenses already paid, or which cannot be changed by the investment decision
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- They do not affect incremental cash flows, so they do not affect the NPV of a decision. Ignore them!
- Many sunk costs are "related" to a project, but no longer incremental if the current decision does not affect them



Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of future operating cash flows of $1.5 million

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- Launching requires another $1 million investment (marketing and so on), with a present value of future operating cash flows of $1.5 million
- Is this project a good idea today?
- Was this project a good idea last year?

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- Recognize and include the **opportunity cost** of every resource, the value of its "best alternative use"
- E.g., you plan a new factory building that costs $10m to construct, on a piece of land you already own (and do not need for anything else). Alternatively, you could sell the land for $3m
- What is the actual cost of building the factory?

Rule d, Overhead

- **Overhead**, ongoing administrative expenses of a business that cannot be attributed to any specific business activity, but are still necessary for the business to function
  - E.g., rent and utilities of the central office, top management salaries

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- The principle of incremental cash flows says only *changes* in overhead that result from a project count

Rule e, Working Capital

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  - Easy to forget! (The data center example had them)

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- Working capital often keeps changing during the project, and is often recovered at the end
- Common mistakes,
  1. Forgetting about working capital entirely
  2. Forgetting that working capital may change during the life of the project
  3. Forgetting that working capital is often recovered at the end of the project

Rule f, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows
  - E.g., selling the machines and equipment of the project, a positive cash flow!

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  - E.g., selling the machines and equipment of the project, a positive cash flow!
- E.g., in the data center example, suppose it could be sold for \$500 million at $t = 5$, a terminal cash flow of 500
- Discounted, it adds $500/1.08^5 = \$340.29$ million, and now the NPV is positive, $-107.23 + 340.29 = +\$233.06$ million

Example, the Data Center with a Terminal Cash Flow

- Solution ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
Depreciation		200	200	200	200	200
EBIT		40	40	40	40	40
Net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		200	200	200	200	200
Operating cash flows		231.6	231.6	231.6	231.6	231.6
Capital investment	−1,000					
Sale of the data center						500
Working capital	−100					100
Total cash flows	−1,100	231.6	231.6	231.6	231.6	831.6
NPV	233.06					
IRR	14.2%					

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- Terminal cash flows can also be negative, e.g., if we have to pay to shut down the project



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- E.g., oil wells require "capping" when they no longer produce. Capping is expensive, so oil well projects are worth more if you do not cap them (although this is illegal!)
- But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



Practice Problem

- A firm plans a new manufacturing plant on a site it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. The market value of the site
b. The market value of the existing buildings
c. Demolition costs and site clearance
d. The cost of a new access road put in last year
e. Lost cash flows on other projects due to executive time spent on the new facility
f. Future depreciation of the new plant
g. The reduction in the firm's tax bill resulting from depreciation of the new plant
h. The initial investment in inventories of raw materials
i. Money already spent on engineering design of the new plant

Practice Problem

a, b. The site and buildings could have been sold or put to another use. Their values are opportunity costs, treated as incremental cash outflows

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h. The initial investment in inventories is a cash outflow

i. The expenditure on design is a sunk cost and not an incremental cash flow

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- Count indirect effects, opportunity costs, working capital, and terminal cash flows, ignore sunk costs and allocated overhead

Module 7, Part 4

Full-Scale Capital Budgeting Problems

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - It costs 1,000 and lasts 5 years, depreciated straight-line

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- This time, build the whole table in one go

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- Solution ($ millions), the operating cash flows first, then working capital, then the capital investment, then the totals

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Operating cash flows		231.6	231.6	231.6	231.6	231.6
Working capital	−100					100
Capital investment	−1,000					
Total cash flows	−1,100	231.6	231.6	231.6	231.6	331.6
NPV	−107.23					

- NPV = −$107.23 million, reject the project, the same answer as in Part 2

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3. Add the project's capital investment at the start and the terminal cash flows at the end
4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

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- The initial investment in machinery is $6 million. The equipment is depreciated straight-line over 6 years to zero

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- The initial investment in machinery is \$6 million. The equipment is depreciated straight-line over 6 years to zero
- \$250,000 in working capital is required, all purchased immediately and recovered at the end
- Sales are projected at 1 million bottles in the first year, growing at 5% a year over the six years. Production costs are \$2.00 per bottle, and each bottle sells for \$4.00

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- The new design is phased out after 6 years. The tax rate is 21%, and the required rate of return is 12%
- What is the project's NPV?

Example 1, Growing Sales

- Operating cash flows ($ thousands),

Year	1	2	3	4	5	6
Sales (thousand bottles)	1,000	1,050	1,103	1,158	1,216	1,276
Sales ($)	4,000	4,200	4,410	4,631	4,862	5,105
Cost ($)	2,000	2,100	2,205	2,315	2,431	2,553
Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
EBIT	1,000	1,100	1,205	1,315	1,431	1,553
Taxes	210	231	253	276	301	326
Net income	790	869	952	1,039	1,130	1,227
+ Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
Operating cash flows	1,790	1,869	1,952	2,039	2,130	2,227

Example 1, Growing Sales

- Total cash flows ($ thousands),

Year	0	1	2	3	4	5	6
Capital	−6,000						
Working capital	−250						250
Operating cash flows		1,790	1,869	1,952	2,039	2,130	2,227
Total cash flows	−6,250	1,790	1,869	1,952	2,039	2,130	2,477
PV	−6,250	1,598	1,490	1,389	1,296	1,209	1,255
NPV	1,987						

Practice Problem

- A firm has developed a new mousetrap. It can go into production for an initial investment in equipment of $6.3 million. The equipment will be depreciated straight-line over 6 years
- Working capital at each date must be maintained at 10% of next year's forecast sales. Production costs are $1.80 per trap and the traps sell for $7 each. The project ends in 6 years when the trap becomes obsolete. The tax rate is 40%, and the required rate of return is 11%

Year	0	1	2	3	4	5	6	After
Sales (million traps)	0	0.6	0.8	0.9	0.9	0.5	0.2	0

Practice Problem

- Suppose the firm can cut its working capital requirements in half with better inventory control. By how much does this increase the project NPV?

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- Suppose the firm can cut its working capital requirements in half with better inventory control. By how much does this increase the project NPV?
- Solution ($ millions), working capital at each date is 10% of next year's sales,

Year	0	1	2	3	4	5	6
Sales units		0.600	0.800	0.900	0.900	0.500	0.200
Revenues	0.000	4.200	5.600	6.300	6.300	3.500	1.400
NWC	0.420	0.560	0.630	0.630	0.350	0.140	0.000
Cash flow from NWC	−0.420	−0.140	−0.070	0.000	0.280	0.210	0.140

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

- Discounting the changes in cash flow at 11%, the change in NPV is $0.1095 million

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- Use cash flows, not net income, treat financing separately, and remember the depreciation tax shield
- Only incremental cash flows count, include indirect effects, opportunity costs, working capital, and terminal cash flows, ignore sunk costs
- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

Thank you!

Please let me know if you have any questions.